



Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : SCMS - Student Course Management System

Group No : 03, CSE370 Lab Section : 10, Spring 2026

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Introduction

Our project is a **Student Course Management System (SCMS)**. It is a web application that helps students plan their courses, avoid schedule conflicts, calculate CGPA, track academic tasks, find available rooms, share course resources, and book consultations with faculty. Faculty members can manage their advisee's, consultation slots, and course resources. The main goal of SCMS is to make academic planning easier for students while allowing faculty to manage their tasks efficiently.

Project Features

1. Intelligent Course Prerequisite Planner

Students input their current semester and completed courses. The system validates prerequisite requirements and generates a curated list of eligible courses for the upcoming semester, ensuring students pursue courses they are academically qualified to take.

2. Conflict-Free Schedule Generator

When students select specific course sections, the system cross-references class timings and notifies them of any scheduling conflicts. This prevents overlaps and ensures students can attend all selected courses without conflicts.

3. Dynamic CGPA & Target Grade Predictor

A sophisticated calculator that tracks historical academic performance. Students can set a target CGPA, and the system computes the exact grades required in current courses to achieve that goal, providing clarity on academic objectives and required effort.

4. Interactive Course Planner

Students select courses for the upcoming semester through an intuitive interface. The system calculates total credit hours and enforces credit-load limitations to prevent registration overload and maintain academic sustainability.

5. Academic Task Tracker

A Kanban-style task management board for organizing assignments and exams. The system automatically links tasks to enrolled courses and sorts them by due date, keeping students focused on immediate priorities.

6. Room & Lab Availability Finder

Students can search for unoccupied study spaces in real-time. The system queries the schedule database against the current time or a specified time block, delivering availability information suitable for group study or independent work.

7. Course Resource Hub

A centralized repository for student-shared academic resources, including notes, video tutorials, and study guides organized by course. An upvote/downvote mechanism ensures highly-rated resources appear at the top.

8. Faculty Consultation Booking

A directory displaying faculty office hours with an integrated booking system. Students can reserve consultation slots using a calendar interface while preventing double-booking conflicts.

9. Faculty & Advisor Portal

Faculty members can log in to confirm or deny consultation requests and serve as academic advisors, guiding students in course selection and academic planning.

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Features [3 per member]**

Ft 1 Intelligent Course

Prerequisite Planner Ft 2

Conflict-Free Schedule Generator

Ft 3 Dynamic CGPA & Target

Grade Predictor Ft 1 Interactive

Course Planner

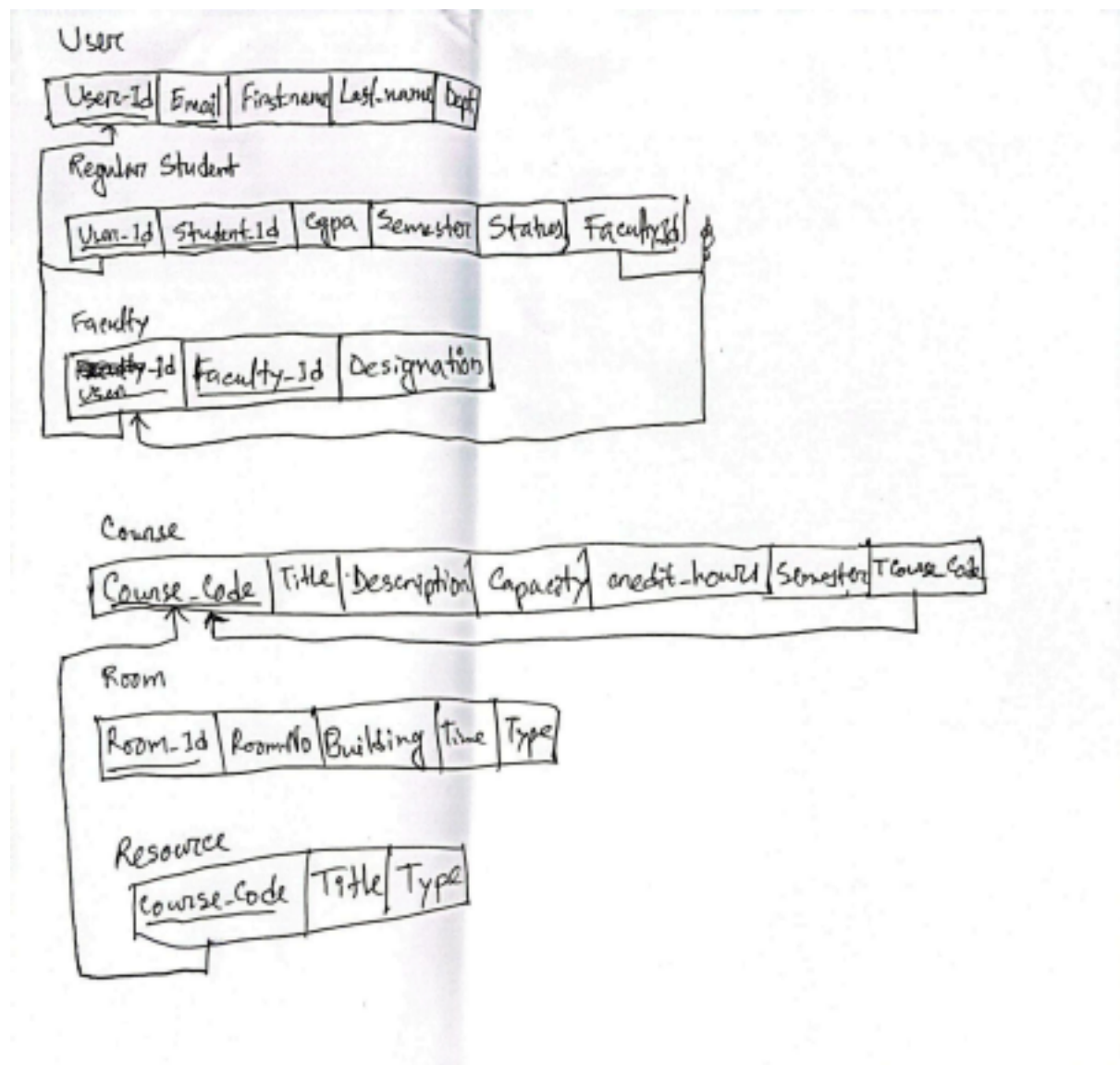
Ft 2 Academic Task Tracker

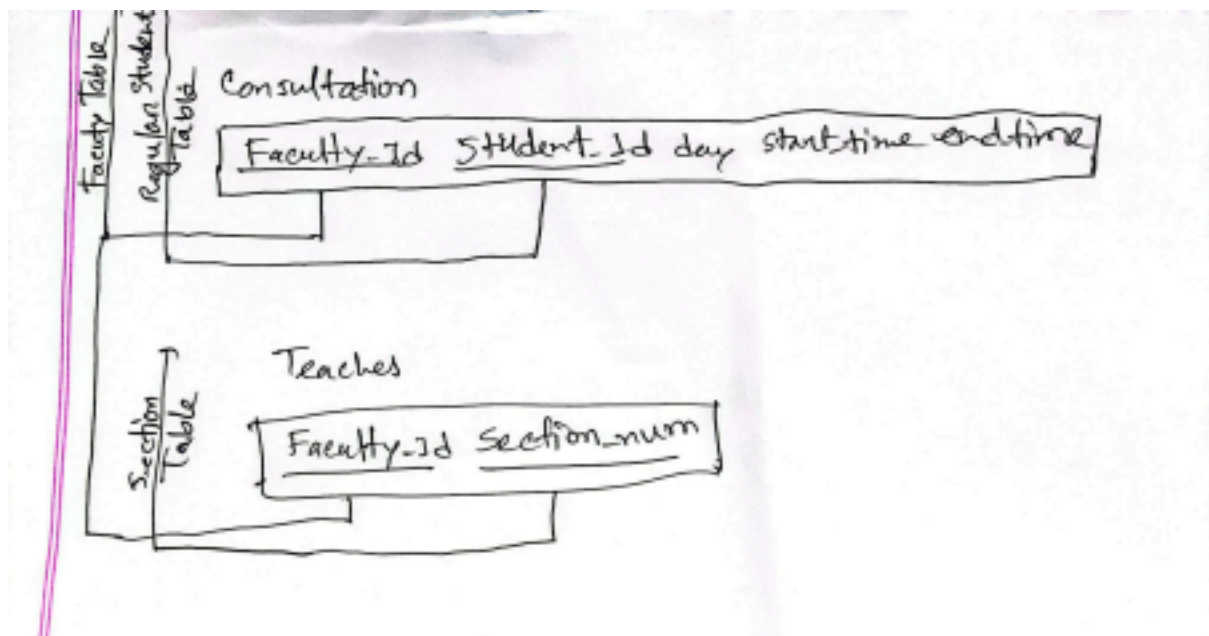
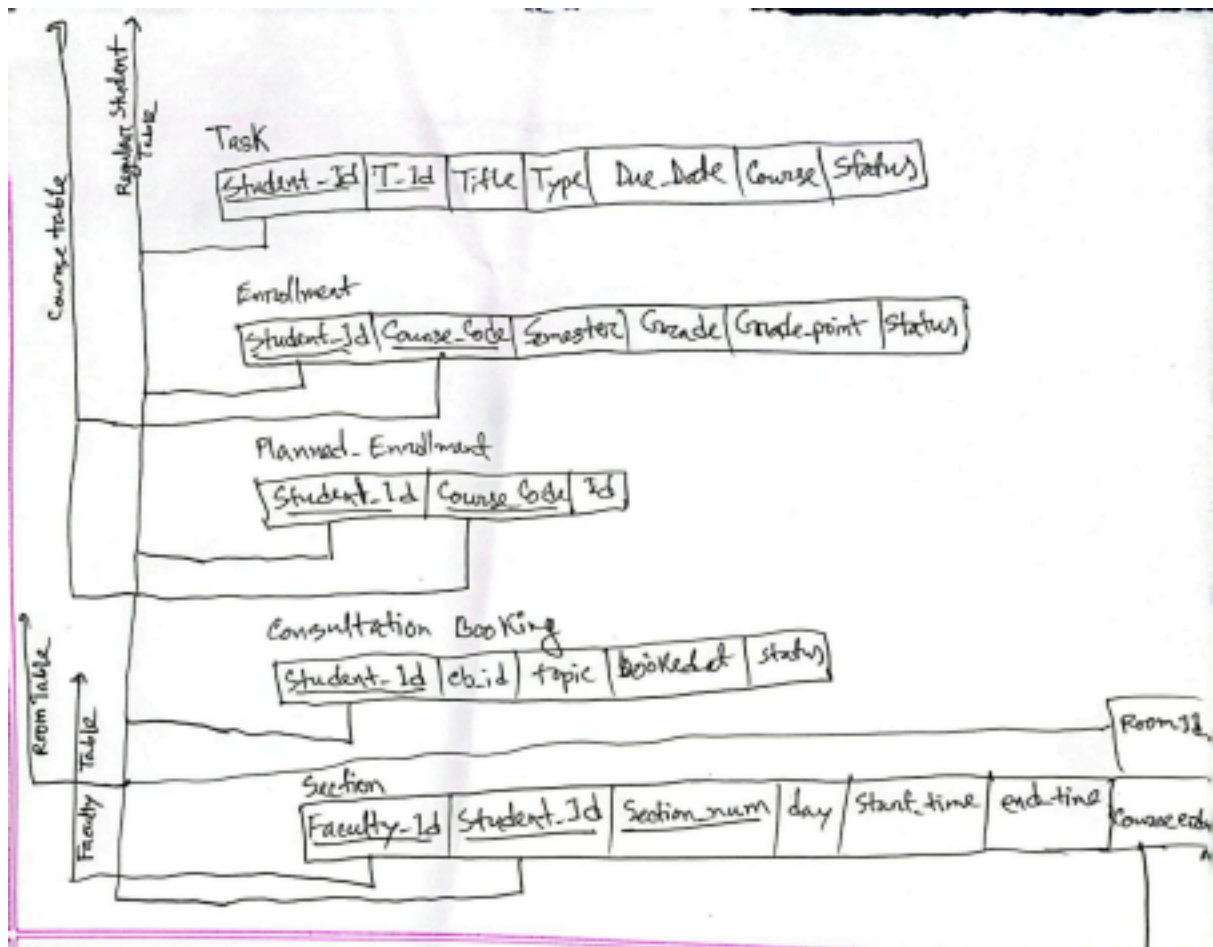
Ft 3 Room & Lab Availability

Finder Ft 1 Course Resource Hub

Ft 2 Faculty Consultation Booking

Ft 3 Faculty & Advisor Portal





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Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF. **Ans:** Yes, It is in 1NF form. Because there is not multivalued, composite and nested attribute in the schema table.

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema? **Ans:** Yes, It is in 2NF form. Because, It is in 1NF form & no functional dependencies appear.
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema? **Ans:** Yes, It is in 3NF form. Because It is in 2NF form & no transient dependencies appear.

The frontend of our project was built using **React**, **Vite**, **Tailwind CSS**, and **shadcn/ui components**. The user interface is responsive and interactive. Students can view eligible

courses, track CGPA, manage tasks, check room availability, and book consultations. Faculty members can view advisee's, manage consultations, and post course resources.

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The backend is built using **plain PHP API files** located in src/apis and communicates with a

MySQL/MariaDB database named scms. Each PHP file represents an API endpoint that handles requests like fetching courses, enrolling students, managing tasks, and booking consultations. The React frontend calls these APIs using fetch.

Key backend functionalities include:

1. **Course Prerequisite Checker:** Validates if a student can enroll in a course based on completed prerequisites.
2. **Conflict-Free Schedule Generator:** Ensures that course sections do not overlap.
3. **CGPA and Target Grade Predictor:** Calculates current CGPA and predicts grades needed to achieve a target CGPA.
4. **Task Tracker Management:** Manages academic tasks linked to courses.
5. **Room and Lab Finder:** Shows free rooms and labs based on student inputs.
6. **Resource Hub:** Manages course materials and rankings based on upvotes and downvotes.
7. **Faculty Consultation Booking:** Allows students to book consultation slots with faculty members.

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Intelligent Course Prerequisite Planner

Conflict-Free Schedule Generator

Dynamic CGPA & Target Grade Predictor

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Interactive Course Planner

Academic Task Tracker

Room & Lab Availability Finder

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Course Resource Hub

Faculty Consultation Booking

Faculty & Advisor Portal

Conclusion

The SCMS project successfully integrates frontend and backend systems to provide a complete academic planning tool. Students can plan courses, track grades, manage tasks, and interact with faculty efficiently. Faculty can oversee advisee's, manage resources, and schedule consultations smoothly.

References

N/A